THM LAB COMPLETION REPORT

Active Reconnaissance

Name: Muhammad Afnan K O

Focus: WAPT

Platform: TryHackMe.com

1. Objective — What was the task about?

The main goal of this lab was to learn how to gather information by directly interacting with a target. Unlike passive recon, the objective here was to actively probe the target's network and servers to find open ports, running services, and hidden web directories in order to find the lab's hidden flags

2. Concept/Understanding — What did you learn?

I learned how active reconnaissance works and why it is different from passive methods. Active recon requires sending actual requests or ping to the target to see how it responds. I know about pings and requests but in this class I understood what is it used for, it also noisy or it means the server logs our requests and they can see our requests and can easily be picked up by firewalls or intrusion detection systems.

3. Practical Work — What did you perform?

I performed some scans using nmap and other scanning tools to check if the server is up and check how many hosts are up in a network. Also used traceroute to check the paths the requests are travelled , which maps the network of request. And I used gobuster and dirb for bruteforcing the directories in the server which give the active file directories and hidden ones, and telenet and netcat helped to listen network requests in a server.

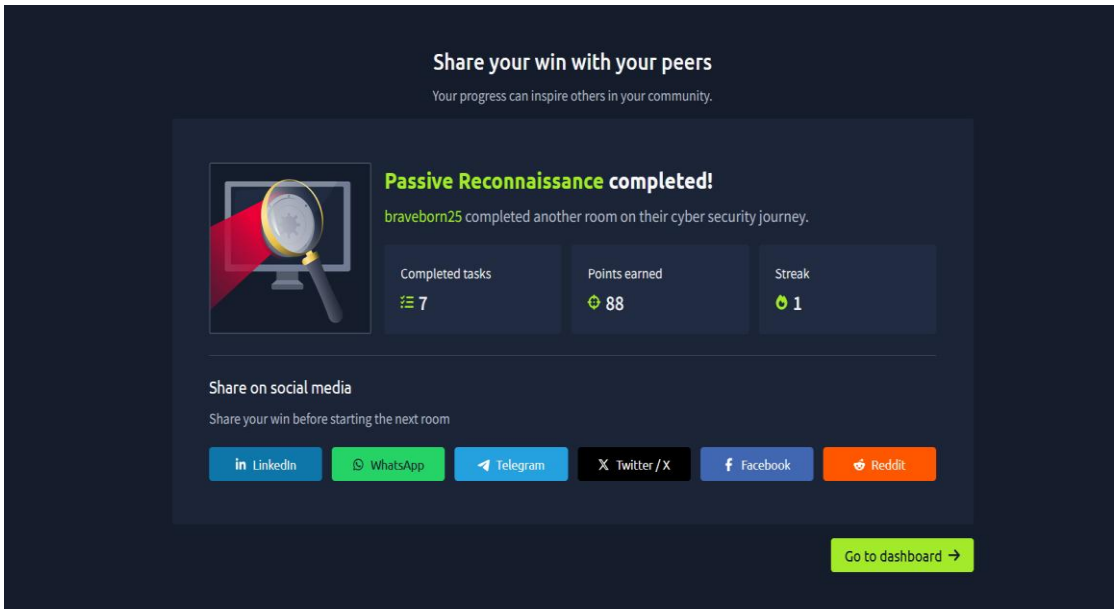
4. Tools Used — Tools used during the task.

- Nmap
- Traceroute
- Gobuster/Dirb
- Telnet/Netcat

5. Evidence — Relevant screenshots and results.

Screenshots of the completed TryHackMe rooms, practical exercises, and relevant results can be added below as evidence of the work completed.

1.Screenshot of room completion:-



2.Screenshot of Ping operation:-

```
user@AttackBox$ ping -c 5 MACHINE_IP
PING MACHINE_IP (MACHINE_IP) 56(84) bytes of data.
64 bytes from MACHINE_IP: icmp_seq=1 ttl=64 time=0.512 ms
64 bytes from MACHINE_IP: icmp_seq=2 ttl=64 time=0.478 ms
64 bytes from MACHINE_IP: icmp_seq=3 ttl=64 time=0.491 ms
64 bytes from MACHINE_IP: icmp_seq=4 ttl=64 time=0.503 ms
64 bytes from MACHINE_IP: icmp_seq=5 ttl=64 time=0.485 ms

--- MACHINE_IP ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4098ms
rtt min/avg/max/mdev = 0.478/0.494/0.512/0.012 ms
```

3.Screenshot of Traceroute:-

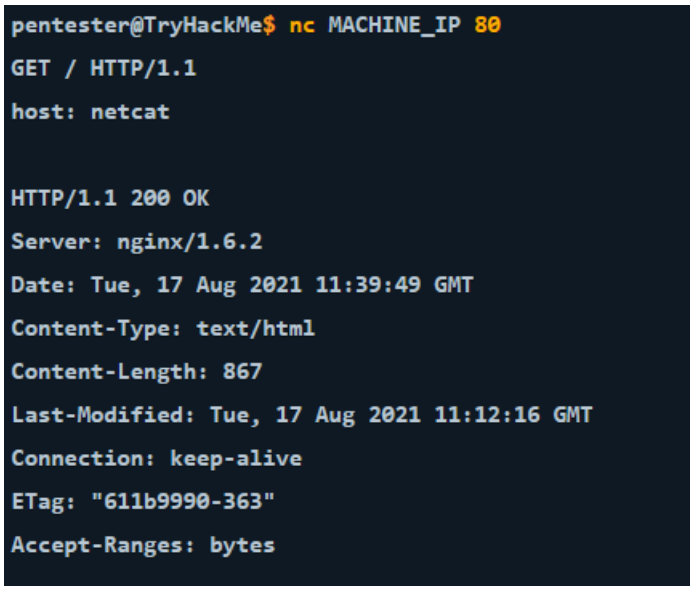
```
user@AttackBox$ traceroute tryhackme.com
traceroute to tryhackme.com (172.67.69.208), 30 hops max, 60 byte packets
 1 ec2-3-248-240-5.eu-west-1.compute.amazonaws.com (3.248.240.5) 2.663 ms * ec2-3-248-240-13.eu-west-1.compute.amazonaws.com
 2 100.66.8.86 (100.66.8.86) 43.231 ms 100.65.21.64 (100.65.21.64) 18.886 ms 100.65.22.160 (100.65.22.160) 14.556 ms
 3 * 100.66.16.176 (100.66.16.176) 8.006 ms *
 4 100.66.11.34 (100.66.11.34) 17.401 ms 100.66.10.14 (100.66.10.14) 23.614 ms 100.66.19.236 (100.66.19.236) 17.524 ms
 5 100.66.7.35 (100.66.7.35) 12.808 ms 100.66.6.109 (100.66.6.109) 14.791 ms *
 6 100.65.14.131 (100.65.14.131) 1.026 ms 100.66.5.189 (100.66.5.189) 19.246 ms 100.66.5.243 (100.66.5.243) 19.805 ms
 7 100.65.13.143 (100.65.13.143) 14.254 ms 100.95.18.131 (100.95.18.131) 0.944 ms 100.95.18.129 (100.95.18.129) 0.778 ms
 8 100.95.2.143 (100.95.2.143) 0.680 ms 100.100.4.46 (100.100.4.46) 1.392 ms 100.95.18.143 (100.95.18.143) 0.878 ms
 9 100.100.20.76 (100.100.20.76) 7.819 ms 100.92.11.36 (100.92.11.36) 18.669 ms 100.100.20.26 (100.100.20.26) 0.842 ms
10 100.92.11.112 (100.92.11.112) 17.852 ms * 100.92.11.158 (100.92.11.158) 16.687 ms
11 100.92.211.82 (100.92.211.82) 19.713 ms 100.92.0.126 (100.92.0.126) 18.603 ms 52.93.112.182 (52.93.112.182) 17.738 ms
12 99.83.69.207 (99.83.69.207) 17.603 ms 15.827 ms 17.351 ms
13 100.92.9.83 (100.92.9.83) 17.894 ms 100.92.79.136 (100.92.79.136) 21.250 ms 100.92.9.118 (100.92.9.118) 18.166 ms
14 172.67.69.208 (172.67.69.208) 17.976 ms 16.945 ms 100.92.9.3 (100.92.9.3) 17.709 ms
```

4.Screenshot of Telnet:-

```
pentester@TryHackMe$ telnet MACHINE_IP 80
Trying MACHINE_IP...
Connected to MACHINE_IP.
Escape character is '^]'.
GET / HTTP/1.1
host: telnet

HTTP/1.1 200 OK
Server: nginx/1.6.2
Date: Tue, 17 Aug 2021 11:13:25 GMT
Content-Type: text/html
Content-Length: 867
Last-Modified: Tue, 17 Aug 2021 11:12:16 GMT
Connection: keep-alive
ETag: "611b9990-363"
Accept-Ranges: bytes
...
```

5.Screenshot of netcat :-



6. Result — What did you achieve or discover?

I successfully completed the Active Reconnaissance lab. I was able to map out the target's open ports, identify the exact versions of the services running, and discover hidden web directories. This gave me a very clear, hands-on understanding of how attackers map out a target's attack surface by actively probing it.

7. Key Learnings — What did you learn from the task?

- Active scanning provides essential details (like software versions) that you usually can't get passively.
- Nmap is an incredibly powerful tool for finding open ports on a network.
- Directory busting is a highly effective way to find hidden administrative logins or forgotten backup files on a web server.
- Interacting directly with a target leaves a footprint, so understanding how much noise a scan makes is important.

Conclusion

The Active Reconnaissance lab was a great step up from passive footprinting. It gave me real, hands-on experience using command-line tools to actively engage with a target and uncover its actual setup. Learning how to properly scan a network and brute-force web directories built my confidence in discovering the technical vulnerabilities needed for the next steps of a penetration test.